HOT SPOTS AND VERTEX INTERPOLATION ON A REGULAR SIMPLEX IN EVERY DIMENSION

Abstract. Let $T \subset \mathbb{R}^n$ be a regular simplex, where $n \geq 2$. We prove that the eigenspace of its first positive Neumann eigenvalue has dimension n and that evaluation at the $n+1$ vertices identifies this eigenspace with the space of zero-sum real vertex data. Every nonzero first eigenfunction attains its global maximum and minimum exactly at the vertices having the largest and smallest vertex values, respectively. The inequalities are strict at every nonvertex point of the closed simplex, including all lower-dimensional boundary strata. The proof first identifies the eigenspace by an injective mean-gradient map and simplex symmetry. A chamber for the stabilizer of one vertex is an orthoscheme; an absolute-value argument for tangent vector fields gives strict signs for all derivatives in its orthonormal edge coordinates. A local Neumann argument supplies strict comparison along closed segments in every dimension. Normalized vertex modes then give positive interpolation weights, which yield the classification for arbitrary eigenfunctions and every pattern of tied extrema.

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1. Introduction and main results

For a bounded connected Lipschitz domain $P \subset \mathbb{R}^n$, write $0 = \mu_0(P) < \mu_1(P) \leq \mu_2(P) \leq \cdots$ for the Neumann eigenvalues, repeated according to multiplicity. Throughout this paper, *first*

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eigenfunction means an eigenfunction associated with the first *positive* eigenvalue $\mu_1(P)$. Its real eigenspace is

$$E_1(P) = \left\{ u \in H^1(P) : \int_P \nabla u \cdot \nabla \eta \, dx = \mu_1(P) \int_P u\eta \, dx \quad \text{for every } \eta \in H^1(P) \right\}. \tag{1.1}$$

In particular, $\int_P u = 0$ for $u \in E_1(P)$. The hot spots property asks that the global extrema of every nonzero first eigenfunction occur on the boundary. For a regular simplex we establish the more precise statement that all such extrema are vertices, and we determine exactly which vertices occur.

The variational approach to hot spots studies a Laplace operator on tangent vector fields and the effect of replacing gradient components by their absolute values. Kennedy and Rohleder developed this approach in arbitrary dimension for lip domains and suitable symmetric domains [4]. Their geometric condition is that, in orthonormal coordinates, an almost-everywhere defined boundary normal has either one nonzero component or two nonzero components of opposite signs. The chambers used below have exactly this property. The flat-boundary identity underlying the vector argument is related to the Gaffney and Korn identities discussed in [2]. We give the needed polyhedral identity and its equality consequences explicitly.

The spectral and boundary arguments use different regularity inputs. Homogeneous Neumann problems on bounded convex polytopes have H^2 regularity in every dimension [1, Proposition 4.8]. Continuity and positivity at corners will instead follow from a local harmonic lift, boundary flattening, reflection, and the interior elliptic theory [3, Chapter 8]; see also [5, 6]. This separation is necessary when $n \geq 4$, where H^2 alone does not embed into the space of continuous functions.

Let

$$T = \operatorname{int} \operatorname{conv}\{v_1, \dots, v_N\} \subset \mathbb{R}^n, \qquad N = n + 1, \tag{1.2}$$

be a regular simplex. The vertices are labeled but otherwise arbitrary.

Theorem 1.1 (Eigenspace and complete extremal-set classification)**.** *For every regular simplex* (1.2) *with $n \geq 2$, the following assertions hold.*

(i) *Every $u \in E_1(T)$ has a continuous representative on $\overline{T}$, and*

$$\dim E_1(T) = n. \tag{1.3}$$

(ii) *The vertex-evaluation map is a linear isomorphism*

$$\mathrm{ev}_V : E_1(T) \longrightarrow H_0 := \left\{ c \in \mathbb{R}^N : \sum_{i=1}^N c_i = 0 \right\}, \qquad u \longmapsto \big(u(v_1), \dots, u(v_N)\big). \tag{1.4}$$

(iii) *If $0 \neq u \in E_1(T)$ and $c_i = u(v_i)$, then*

$$\arg\max_{\overline{T}} u = \{v_i : c_i = \max_j c_j\}, \tag{1.5}$$

$$\arg\min_{\overline{T}} u = \{v_i : c_i = \min_j c_j\}. \tag{1.6}$$

More precisely, at every nonvertex point $x \in \overline{T}$,

$$\min_i c_i < u(x) < \max_i c_i. \tag{1.7}$$

(iv) *Vertex evaluation is an isometry for the supremum norms:*

$$\|u\|_{L^\infty(T)} = \max_{1 \leq i \leq N} |u(v_i)|. \tag{1.8}$$

The exact descriptions (1.5)–(1.6) include arbitrary ties. In particular, an edge or a positive-dimensional face spanned by vertices of equal maximal value contains no additional maximizer in its relative interior.

The following interpolation statement describes the mechanism behind the classification.

Theorem 1.2 (Positive interpolation weights)**.** *There are unique functions $h_1, \dots, h_N \in \mathbb{R} \oplus E_1(T)$ with*

$$h_i(v_j) = \delta_{ij}. \tag{1.9}$$

They satisfy

$$\sum_{i=1}^{N} h_i = 1 \quad \text{on } \overline{T}, \qquad 0 < h_i(x) < 1 \quad \text{if } x \in \overline{T} \setminus \{v_1, \dots, v_N\}. \tag{1.10}$$

Writing $\mu = \mu_1(T)$, their equations are

$$-\Delta h_i = \mu \left(h_i - \frac{1}{N} \right) \quad \text{in } T, \qquad \partial_\nu h_i = 0 \quad \text{on } \partial T, \tag{1.11}$$

in the weak sense. Every $u \in E_1(T)$ obeys

$$u(x) = \sum_{i=1}^{N} u(v_i) h_i(x), \qquad x \in \overline{T}. \tag{1.12}$$

Moreover, $\psi_i = h_i - 1/N$ belong to $E_1(T)$, span that space, and have the sole linear relation $\sum_i \psi_i = 0$.

The N functions ψ_i are thus generators of an n-dimensional space, and every eigenfunction has a unique zero-sum coefficient vector in this generating family. The functions h_i are affine shifts of eigenfunctions, as expressed by (1.11).

Outline of the proof. Section 2 proves the tangent-field variational identity, its equality characterization, and injectivity of the mean-gradient map on the first eigenspace of a convex polytope. Section 3 establishes continuity at arbitrary boundary strata and a strict comparison lemma for segments in the closed polytope. Section 4 uses simplex symmetry to determine the eigenspace and identifies the chamber of a vertex stabilizer as an orthoscheme. Reflection extension shows that its first positive Neumann eigenvalue equals that of the simplex and is simple. Section 5 proves strict coordinate monotonicity on that chamber and determines all extrema of a vertex mode. Section 6 constructs the positive weights and completes both main theorems. Section 7 describes every extremal pattern and illustrates the conclusions in dimensions two, three, and four. All arguments preceding the illustrations apply to every $n \geq 2$.

2. Tangent vector fields and the first eigenspace

Throughout this section, $P \subset \mathbb{R}^n$ is the interior of a bounded full-dimensional convex polytope, $n \geq 2$, and $\lambda = \mu_1(P)$. The standard variational characterization is

$$\int_P |\nabla w|^2 \, dx \geq \lambda \int_P w^2 \, dx, \qquad w \in H^1(P), \quad \int_P w \, dx = 0. \tag{2.1}$$

Equality holds precisely when $w \in E_1(P)$, including $w = 0$.

2.1. Regularity, density, and the flat-boundary identity.

The homogeneous Neumann realization of $-\Delta$ on a bounded convex domain has operator domain contained in $H^2(P)$ [1, Proposition 4.8]. In particular, the mean-zero weak solution of a compatible Neumann Poisson equation with L^2 right-hand side belongs to $H^2(P)$. Its normal derivative is zero in the trace sense on each relatively open facet. We will use this result both for eigenfunctions and for the auxiliary Poisson equation below.

Define

$$\mathcal{V}(P) = \{X \in H^1(P; \mathbb{R}^n) : X \cdot \nu = 0 \text{ on every relatively open facet of } P\}. \tag{2.2}$$

For a vector field $X = (X_1, \dots, X_n)$, use the Jacobian convention $(DX)_{ij} = \partial_j X_i$ and set

$$|dX|^2 = \sum_{1 \le i < j \le n} (\partial_i X_j - \partial_j X_i)^2. \tag{2.3}$$

The summation convention in (2.3) fixes the normalization of the antisymmetric derivative in every dimension.

Lemma 2.1 (Tangential density and flat identity)**.** *Smooth tangent fields supported away from the faces of codimension at least two are dense in $\mathcal{V}(P)$ for the H^1 norm. For every $X \in \mathcal{V}(P)$,*

$$\int_P |DX|^2 \, dx = \int_P (\operatorname{div} X)^2 \, dx + \int_P |dX|^2 \, dx. \tag{2.4}$$

Proof. Let S be the union of all faces of dimension at most $n-2$, and put $d(x) = \operatorname{dist}(x, S)$. A finite covering of the faces by bounded pieces of affine subspaces gives

$$F(r) := |\{x \in P : d(x) < r\}| \le C r^2 \qquad (0 < r < r_0). \tag{2.5}$$

For $0 < \varepsilon < \min(r_0, 1/2)$, choose the Lipschitz cutoff

$$\chi_\varepsilon(x) = \begin{cases} 0, & d(x) \le \varepsilon^2, \\ \dfrac{\log(d(x)/\varepsilon^2)}{|\log \varepsilon|}, & \varepsilon^2 < d(x) < \varepsilon, \\ 1, & d(x) \ge \varepsilon. \end{cases}$$

Since $|\nabla d| \le 1$ almost everywhere, Stieltjes integration by parts and (2.5) imply

$$\int_{\{\varepsilon^2 < d < \varepsilon\}} d^{-2} \, dx \le C + 2C \log(1/\varepsilon).$$

Consequently,

$$\int_P |\nabla \chi_\varepsilon|^2 \, dx \le \frac{C}{|\log \varepsilon|} \longrightarrow 0. \tag{2.6}$$

For $M > 0$, apply the radial truncation $T_M(z) = z$ for $|z| \le M$ and $T_M(z) = Mz/|z|$ otherwise. The Sobolev chain rule gives

$$\|T_M(X) - X\|_2^2 \le \int_{\{|X| > M\}} |X|^2 \, dx, \qquad \|D(T_M(X) - X)\|_2^2 \le 4 \int_{\{|X| \ge M\}} |DX|^2 \, dx.$$

Both right-hand sides tend to zero as $M \to \infty$, so $T_M(X) \to X$ strongly in $H^1(P)$. The radial truncation preserves tangency, including in the trace sense. For fixed M, the product rule gives

$$D((1 - \chi_\varepsilon) T_M(X)) = (1 - \chi_\varepsilon) D(T_M(X)) - T_M(X) \otimes \nabla \chi_\varepsilon.$$

The first term tends to zero in L^2 by dominated convergence; the second has L^2 norm at most $M \|\nabla \chi_\varepsilon\|_2$, which tends to zero by (2.6). The L^2 convergence of the fields themselves also follows by dominated convergence. Thus $\chi_\varepsilon T_M(X) \to T_M(X)$ strongly in $H^1(P)$. It remains to smooth tangent fields which vanish in a neighborhood of S.

Fix such a field. Its support in $\overline{P}$ is compact and disjoint from S. Cover this support by finitely many interior balls and boundary balls, each boundary ball centered on a single facet and meeting no other facet. Choose a smooth partition of unity with each localized field supported compactly inside its ball. In a boundary ball, extend the tangential components evenly and the normal component oddly across the facet hyperplane. The zero trace of the normal component makes its odd extension an H^1 extension. The localized field vanishes near the spherical boundary, so the reflected field can then be extended by zero to $\mathbb{R}^n$ without a jump. Mollification with a radial kernel preserves these parities and hence tangency on the hyperplane. Choose the mollifier radius smaller than the distances of the localized support to the boundary of its ball and to S.

The smoothed field then meets no other facet and still vanishes near S. Interior pieces are zero-extended and mollified in the same way. Restricting to P and summing gives smooth tangent fields converging to the fixed field in $H^1(P)$. Choosing successively $M \to \infty$, $\varepsilon \downarrow 0$, and the mollifier radii tending to zero gives a diagonal sequence converging to the original X. This proves the density assertion.

For a smooth tangent field obtained in this way,

$$\operatorname{div}\{X \operatorname{div} X - (X \cdot \nabla)X\} = (\operatorname{div} X)^2 - \sum_{i,j} \partial_i X_j \, \partial_j X_i,$$
$$|dX|^2 = |DX|^2 - \sum_{i,j} \partial_i X_j \, \partial_j X_i.$$

The first vector field has zero normal flux on each facet: its first term does because $X \cdot \nu = 0$, and its second does because the normal is constant and

$$\nu \cdot (X \cdot \nabla)X = X \cdot \nabla(X \cdot \nu) = 0.$$

Here X differentiates tangentially along the facet, where its normal trace vanishes. Integration proves (2.4) for the smooth approximants. Strong H^1 convergence gives the identity for X. □

Corollary 2.2 (Hessian identity)**.** *If $u \in H^2(P)$ has homogeneous Neumann data on every facet, then*

$$\int_P |D^2 u|^2 \, dx = \int_P (\Delta u)^2 \, dx. \tag{2.7}$$

For $u \in E_1(P)$ this becomes

$$\|D^2 u\|_2^2 = \lambda^2 \|u\|_2^2 = \lambda \|\nabla u\|_2^2. \tag{2.8}$$

Proof. Apply (2.4) to $X = \nabla u$, which lies in $\mathcal{V}(P)$ and has $dX = 0$. The eigenfunction identity also uses $-\Delta u = \lambda u$ and $\|\nabla u\|_2^2 = \lambda \|u\|_2^2$. □

Lemma 2.3 (Uniqueness from a facet)**.** *Let z be a weak Neumann eigenfunction on P, with any eigenvalue. If its Dirichlet trace vanishes on a nonempty relatively open facet patch, then $z = 0$ throughout P. Consequently, if $Z \in \mathcal{V}(P)$ and all components of Z satisfy the weak Neumann eigenvalue equation for one common eigenvalue, then $Z = 0$.*

Proof. Choose a point q in the relatively open facet patch and a ball $B = B_r(q)$ small enough that $\partial P \cap B$ is contained in that patch and $P \cap B$ is a half-ball. The zero Dirichlet trace permits extension of z by zero to a function $\widetilde{z} \in H^1(B)$. For every $\eta \in C_c^\infty(B)$, its restriction to $P \cap B$, extended by zero outside B within P, is an admissible $H^1(P)$ test. If γ is the eigenvalue of z, its weak equation gives

$$\int_B \nabla \widetilde{z} \cdot \nabla \eta \, dx = \int_{P \cap B} \nabla z \cdot \nabla \eta \, dx = \gamma \int_{P \cap B} z \eta \, dx = \gamma \int_B \widetilde{z} \eta \, dx.$$

Thus $\widetilde{z}$ solves the same Helmholtz equation across the flat interface and is analytic by constant-coefficient interior elliptic regularity. Since it vanishes on the exterior half-ball, it vanishes on B. The identity theorem for the analytic function z on connected P gives $z = 0$ throughout P.

For the vector assertion, each facet normal ν gives a scalar Neumann eigenfunction $Z \cdot \nu$ whose Dirichlet trace vanishes on that facet. The scalar assertion gives $Z \cdot \nu = 0$ throughout P. The facet normals span $\mathbb{R}^n$: otherwise a nonzero direction orthogonal to all of them would preserve every defining half-space under arbitrary translations, contradicting boundedness. Hence $Z = 0$. □

2.2. **The vector minimum and its equality space.**

Proposition 2.4 (Vector variational principle)**.** *For every $X \in \mathcal{V}(P)$,*

$$\|DX\|_2^2 \geq \lambda \|X\|_2^2. \tag{2.9}$$

Equality holds if and only if $X = \nabla u$ for some $u \in E_1(P)$.

Proof. Since $X \cdot \nu = 0$, the divergence theorem gives $\int_P \operatorname{div} X = 0$. Let $v \in H^2(P)$ be the mean-zero solution of

$$\Delta v = \operatorname{div} X, \qquad \partial_\nu v = 0. \tag{2.10}$$

Existence follows from the scalar Poincaré inequality and the Lax–Milgram theorem on mean-zero $H^1(P)$; the stated H^2 regularity then applies. Its weak formulation is

$$\int_P \nabla v \cdot \nabla \eta \, dx = -\int_P (\operatorname{div} X)\eta \, dx = \int_P X \cdot \nabla \eta \, dx \qquad (\eta \in H^1(P)),$$

where the last equality uses the zero normal trace of X. Set $Y = X - \nabla v$. Then $Y \in \mathcal{V}(P)$, $\operatorname{div} Y = 0$, and

$$\int_P Y \cdot \nabla \eta \, dx = 0 \quad (\eta \in H^1(P)), \qquad \|X\|_2^2 = \|\nabla v\|_2^2 + \|Y\|_2^2. \tag{2.11}$$

The first identity is the weak equation for v just displayed; testing it with $\eta = v$ gives the orthogonality used in the second. Testing with $\eta = x_j$ gives $\int_P Y_j = 0$ for every component. By Lemma 2.1, $d\nabla v = 0$, and $\operatorname{div} Y = 0$,

$$\|DX\|_2^2 = \|\Delta v\|_2^2 + \|dY\|_2^2 = \|\Delta v\|_2^2 + \|DY\|_2^2. \tag{2.12}$$

Integration by parts, Cauchy–Schwarz, and (2.1) give

$$\|\nabla v\|_2^2 = -\int_P v \Delta v \, dx \leq \|v\|_2 \|\Delta v\|_2 \leq \lambda^{-1/2} \|\nabla v\|_2 \|\Delta v\|_2.$$

If $v \neq 0$, its mean-zero normalization and connectedness of P give $\|\nabla v\|_2 > 0$, so division and squaring yield

$$\|\Delta v\|_2^2 \geq \lambda \|\nabla v\|_2^2, \tag{2.13}$$

and the case $v = 0$ is immediate. Equality for nonzero v forces equality in (2.1), hence $v \in E_1(P)$; conversely, $-\Delta v = \lambda v$ gives equality for every $v \in E_1(P)$. Applying (2.1) to every Y_j gives $\|DY\|_2^2 \geq \lambda \|Y\|_2^2$. Equations (2.11)–(2.13) prove (2.9).

If equality holds, equality occurs in each nonnegative scalar defect. Thus $v \in E_1(P)$ and every $Y_j \in E_1(P)$, with zero components allowed. Since $Y \in \mathcal{V}(P)$, the vector assertion of Lemma 2.3 gives $Y = 0$ and $X = \nabla v$. Conversely, gradients of first eigenfunctions attain equality by (2.8). □

Proposition 2.5 (Injective mean-gradient map)**.** *The map*

$$J_P : E_1(P) \longrightarrow \mathbb{R}^n, \qquad J_P(u) = \int_P \nabla u \, dx, \tag{2.14}$$

is injective. In particular, $\dim E_1(P) \leq n$.

Proof. Suppose $J_P(u) = 0$. Each $\partial_j u \in H^1(P)$ has zero mean. Summing its scalar Poincaré inequality and using (2.8) gives

$$\lambda \|\nabla u\|_2^2 \leq \sum_j \|\nabla \partial_j u\|_2^2 = \|D^2 u\|_2^2 = \lambda \|\nabla u\|_2^2.$$

Equality therefore holds for every component. It follows from the variational equality case that $\partial_j u \in E_1(P)$ for every j. This is the step that supplies the Neumann equations for these derivatives.

Since $\nabla u \in \mathcal{V}(P)$ and all its components belong to $E_1(P)$, the vector assertion of Lemma 2.3 gives $\nabla u = 0$. Connectedness and the mean-zero condition then give $u = 0$. □

3. Boundary regularity and strict comparison on closed segments

We retain the convex polytope $P \subset \mathbb{R}^n$. A local weak Neumann equation near $q \in \overline{P}$ means that the weak equation is valid for H^1 test functions whose support stays inside the chosen neighborhood of q, while their traces on the physical boundary of P are unrestricted. No condition is imposed on an artificial spherical boundary.

Lemma 3.1 (Local continuity and positivity)**.** *Suppose $g \in H^1(P \cap B_\rho(q))$ solves the local weak Neumann equation*

$$-\Delta g = \mu g, \qquad \mu \geq 0. \tag{3.1}$$

Then g has a continuous representative on $\overline{P} \cap B_r(q)$ for some $r > 0$. If $g > 0$ in $P \cap B_\rho(q)$, then $g(q) > 0$, and g has a positive lower bound in a smaller relative neighborhood of q.

Proof. An interior point is covered by the usual interior elliptic theory. At a boundary point, form the harmonic lift

$$G(x, \tau) = e^{\sqrt{\mu}\,\tau} g(x) \tag{3.2}$$

on a local product with an open interval I containing zero in the additional variable τ. This lift is in H^1 locally. For a compactly supported test function $\zeta(x, \tau)$ which may meet the lateral physical boundary, the weak equation for g, applied on each τ slice, gives

$$\int \nabla_x G \cdot \nabla_x \zeta = \mu \int G\zeta.$$

Integration by parts in τ gives $\int \partial_\tau G\, \partial_\tau \zeta = -\mu \int G\zeta$. Hence G is weakly harmonic in $n + 1$ variables and preserves the lateral homogeneous Neumann condition.

Translate q to the origin and choose a unit direction strictly inward to every facet incident to q. Such a direction is obtained by pointing toward any fixed interior point of P. A sufficiently small neighborhood meets only incident facets. In coordinates $x = (y, t) \in \mathbb{R}^{n-1} \times \mathbb{R}$, the intersection of their half-spaces has the form

$$t > \varphi(y), \tag{3.3}$$

where φ is the maximum of finitely many affine functions, and is therefore Lipschitz. Flatten the graph by the bi-Lipschitz map

$$\Phi(y, s) = (y, \varphi(y) + s), \qquad s > 0,$$

whose Jacobian determinant is one almost everywhere. Put $H(y, s, \tau) = G(\Phi(y, s), \tau)$. Its weak equation has coefficient matrix

$$B_+ = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} I_{n-1} & -\nabla\varphi \\ -\nabla\varphi^T & 1 + |\nabla\varphi|^2 \end{pmatrix} = D\Phi^{-1} D\Phi^{-T}. \tag{3.4}$$

Since $\nabla\varphi$ is bounded, B_+ is bounded, symmetric, and uniformly elliptic: for $\xi \in \mathbb{R}^{n-1}$ and $\sigma, \theta \in \mathbb{R}$,

$$(\xi, \sigma, \theta)^T B_+ (\xi, \sigma, \theta) = |\xi - \sigma\nabla\varphi|^2 + \sigma^2 + \theta^2,$$

which is comparable to $|\xi|^2 + \sigma^2 + \theta^2$ with constants depending only on $\|\nabla\varphi\|_\infty$. The weak equation for H allows test functions with nonzero trace at $s = 0$.

Choose a box $U \subset \mathbb{R}^{n-1}$ about zero and $a > 0$, and shrink I if necessary, so that the upper half $Q_+ = Q \cap \{s > 0\}$ of the product box $Q = U \times (-a, a) \times I$ maps into the original product neighborhood under $(y, s, \tau) \mapsto (\Phi(y, s), \tau)$, with its image staying away from the artificial

boundary. On Q define

$$\widetilde{H}(y,s,\tau) = H(y,|s|,\tau), \qquad \widetilde{B}(y,s,\tau) = \begin{cases} B_+(y,s,\tau), & s>0, \\ RB_+(y,-s,\tau)R, & s<0, \end{cases} \tag{3.5}$$

where $R = \operatorname{diag}(I_{n-1}, -1, 1)$. The even extension belongs to $H^1(Q)$: its derivatives in y and τ are the even extensions of those of H, while

$$\partial_s \widetilde{H}(y,s,\tau) = \operatorname{sgn}(s)(\partial_s H)(y,|s|,\tau) \quad \text{almost everywhere.}$$

The matching traces at $s=0$ produce no distributional jump. To test its equation, split the integral into $s>0$ and $s<0$ and change s to $-s$ in the latter. The resulting upper-half test is $\zeta(y,s,\tau)+\zeta(y,-s,\tau)$; its admissibility is precisely the weak Neumann condition. Therefore

$$\int_Q \widetilde{B}\nabla\widetilde{H}\cdot\nabla\zeta = 0 \tag{3.6}$$

for all $\zeta \in C_c^\infty(Q)$. Reflection preserves the ellipticity bounds.

Interior Hölder regularity for uniformly elliptic divergence-form equations with bounded measurable coefficients now gives a continuous representative of $\widetilde{H}$ [3, Chapter 8]. By Fubini's theorem, for almost every (y,s) with $s>0$ the continuous function $\tau \mapsto \widetilde{H}(y,s,\tau)$ agrees almost everywhere with $e^{\sqrt{\mu}\,\tau} g(\Phi(y,s))$. Continuity in τ makes this equality valid also at $\tau = 0$. Thus the slice $\widetilde{H}(y,s,0)$, composed with Φ^{-1} and restricted to $s \geq 0$, gives a continuous representative of g up to the physical boundary near q. If $g>0$ in the original interior, $\widetilde{H}$ is nonnegative and nonzero in every ball about $(0,0,0)$ contained in Q. Choose $b>0$ with $\overline{B_{2b}(0)} \subset Q$. The interior Harnack inequality [5] gives

$$0 < \sup_{B_b(0)} \widetilde{H} \leq C \inf_{B_b(0)} \widetilde{H}.$$

In particular, $g(q) = \widetilde{H}(0,0,0) > 0$. Continuity gives a positive lower bound in a smaller neighborhood, and the same holds for g. This proves the lemma. □

Corollary 3.2 (Continuity of Neumann eigenfunctions)**.** *Every weak Neumann eigenfunction on P has a continuous representative on $\overline{P}$.*

Proof. Apply Lemma 3.1 at every point of $\overline{P}$. The continuous local representatives agree on overlaps: they agree almost everywhere, hence everywhere, in the interior, and then on the relative boundary by continuity. They therefore give a continuous representative on all of $\overline{P}$. □

Lemma 3.3 (Local tangential derivative equation)**.** *Let $u \in H^2(P)$ be a weak Neumann eigenfunction with eigenvalue $\mu \geq 0$. Let $q \in \overline{P}$, and let $w \in \mathbb{R}^n$ be tangent to every facet incident to q. Then $D_w u$ satisfies the local weak Neumann equation $-\Delta D_w u = \mu D_w u$ near q.*

Proof. Choose $r>0$ so that every nonincident facet's defining affine function is strictly positive throughout $B_{4r}(q)$. For $|t||w| < r$, translations by tw preserve every incident half-space inequality. In particular,

$$x \in P \cap B_{2r}(q) \quad \Longrightarrow \quad x + tw \in P \cap B_{3r}(q).$$

Define

$$u_t(x) = \frac{u(x+tw)-u(x)}{t} \quad \text{on } P \cap B_{2r}(q), \qquad t \neq 0.$$

Take an H^1 test η supported compactly inside $B_{2r}(q)$, with arbitrary physical boundary trace. The function $x \mapsto \eta(x-tw)$ is an admissible test supported in $B_{3r}(q)$ for the original equation. These tests extend by zero away from their local support to members of $H^1(P)$. On their supports, membership in P is determined only by the incident half-spaces, all of which are invariant under

the translations $\pm tw$. Changing variables in the weak equation tested with $\eta(\,\cdot - tw)$ therefore gives

$$\int_{P\cap B_{2r}(q)} \nabla u(x+tw)\cdot\nabla\eta(x)\,dx = \mu\int_{P\cap B_{2r}(q)} u(x+tw)\eta(x)\,dx.$$

Subtracting the unshifted equation and dividing by t proves

$$\int_{P\cap B_{2r}(q)} \nabla u_t\cdot\nabla\eta = \mu\int_{P\cap B_{2r}(q)} u_t\eta. \tag{3.7}$$

Sobolev absolute continuity on almost every line parallel to w gives

$$u_t(x) = \int_0^1 D_w u(x+\theta tw)\,d\theta,$$
$$\nabla u_t(x) = \int_0^1 D_w\nabla u(x+\theta tw)\,d\theta.$$

All translated segments in these formulas stay in the same local polyhedral domain. To make the convergence explicit, let $F\in L^2(\mathbb{R}^n;\mathbb{R}^{n+1})$ be the zero extension from P of $(D_w u, D_w\nabla u)$, which is square integrable since $u\in H^2(P)$. Minkowski's inequality gives

$$\|u_t - D_w u\|_{H^1(P\cap B_r(q))} \le \int_0^1 \|F(\,\cdot+\theta tw) - F\|_{L^2(\mathbb{R}^n)}\,d\theta \longrightarrow 0.$$

Here the translated points remain in $P\cap B_{2r}(q)$, so the zero extension agrees there with the original derivatives. The limit follows from L^2 translation continuity, uniformly for $0\le\theta\le 1$ as $t\to 0$; no derivative of the zero extension is taken. Passing to the limit in (3.7) for tests supported in $B_r(q)$ proves the asserted local equation. □

Lemma 3.4 (Strict comparison along a closed segment)**.** *Let u be a weak Neumann eigenfunction on P, and let $p_0, p_1\in\overline{P}$ be distinct. Set $w = p_1 - p_0$. If*

$$D_w u > 0 \quad \textit{throughout } P, \tag{3.8}$$

then $u(p_1) > u(p_0)$. If $D_w u < 0$ throughout P, the inequality is reversed.

Proof. Let $q = (p_0+p_1)/2$. Every facet incident to q contains both p_0 and p_1: its nonnegative affine defining function vanishes at their average, so it vanishes at each endpoint. Hence w is tangent to every incident facet. If q is interior the tangency condition is empty.

By Neumann H^2 regularity and Lemma 3.3, $g = D_w u$ satisfies the local weak Neumann equation near q. Its interior positivity and Lemma 3.1 give numbers $c_0, R > 0$ such that $g\ge c_0$ on $P\cap B_R(q)$.

Fix $z\in P$ and, for $0<\varepsilon<1$, put $p_{i,\varepsilon} = (1-\varepsilon)p_i + \varepsilon z$. Every defining affine function of P is strictly positive on the segment joining these points, by convexity and the positive weight of z. Thus that whole segment lies in P. Write its points as

$$x_\varepsilon(s) = (1-\varepsilon)((1-s)p_0 + sp_1) + \varepsilon z, \qquad 0\le s\le 1.$$

Choose $0<\delta<1/2$ and $0<\varepsilon_0<1$ so that $\delta|w| + \varepsilon_0|z-q| < R$. For $|s-1/2|\le\delta$ and $0<\varepsilon<\varepsilon_0$,

$$|x_\varepsilon(s) - q| \le (1-\varepsilon)|s-1/2||w| + \varepsilon|z-q| < R.$$

Consequently $g(x_\varepsilon(s))\ge c_0$ throughout a middle parameter interval of length 2δ, uniformly in such ε. Interior elliptic regularity makes u smooth on the perturbed segment, so the fundamental theorem of calculus gives

$$\begin{aligned} u(p_{1,\varepsilon}) - u(p_{0,\varepsilon}) &= (1-\varepsilon)\int_0^1 g(x_\varepsilon(s))\,ds \\ &\ge 2\delta c_0(1-\varepsilon), \end{aligned}$$

using $g > 0$ on the remainder of the segment. Continuity from Corollary 3.2 permits $\varepsilon \downarrow 0$ and proves $u(p_1) > u(p_0)$. Apply this result to $-u$ for the reversed sign. No derivative of a boundary trace along the original segment is required. □

4. Simplex symmetry and the vertex-stabilizer chamber

We now take T from (1.2), put $N = n+1$, and write $\mu = \mu_1(T)$. A translation and a positive dilation preserve eigenspace dimension and extremal locations under the corresponding change of variables. We therefore normalize the centroid and circumradius by

$$\sum_{i=1}^{N} v_i = 0, \qquad |v_i| = 1, \qquad v_i \cdot v_j = -\frac{1}{n} \quad (i \neq j). \tag{4.1}$$

Every permutation of the vertices is induced by an orthogonal symmetry of T. Indeed, affine independence implies that the vertices span $\mathbb{R}^n$ and that their sole linear relation is $\sum_i v_i = 0$. A vertex permutation preserves this relation and the Gram matrix in (4.1), so it extends to an orthogonal map of $\mathbb{R}^n$ preserving T.

4.1. Dimension and canonical vertex modes.

Proposition 4.1 (The standard simplex representation)**.** *The map*

$$J = J_T : E_1(T) \longrightarrow \mathbb{R}^n, \qquad J(u) = \int_T \nabla u \, dx, \tag{4.2}$$

is a linear isomorphism. In particular, $\dim E_1(T) = n$.

Proof. Injectivity is Proposition 2.5. For an orthogonal symmetry R of T, testing the weak equation for $u \in E_1(T)$ with $\eta \circ R$ and changing variables shows that $u \circ R^{-1} \in E_1(T)$. Its gradient is $R\nabla u(R^{-1}x)$, so the same change of variables gives

$$J(u \circ R^{-1}) = RJ(u). \tag{4.3}$$

Consequently $J(E_1(T))$ is a nonzero invariant subspace under all vertex permutations. It is nonzero because the first eigenspace contains a nonzero function and J is injective.

We check irreducibility directly. If a nonzero invariant subspace contains $a \neq 0$, some difference $v_i - v_j$ has $a \cdot (v_i - v_j) \neq 0$, since the vertex differences span $\mathbb{R}^n$. The transposition interchanging v_i, v_j fixes their perpendicular bisector hyperplane and is the reflection normal to $v_i - v_j$. Subtracting the reflected image of a produces a nonzero multiple of $v_i - v_j$ in the invariant subspace. Its permutation orbit contains all vertex differences and hence spans $\mathbb{R}^n$. Thus $J(E_1(T)) = \mathbb{R}^n$, as required. □

For $a \in \mathbb{R}^n$, define

$$f_a = J^{-1}(a), \qquad F_i = f_{v_i} \quad (1 \leq i \leq N). \tag{4.4}$$

Linearity, (4.1), and (4.3) give

$$f_a(Rx) = f_{R^{-1}a}(x), \qquad \sum_{i=1}^{N} F_i = 0. \tag{4.5}$$

These identities hold on $\overline{T}$ for the continuous representatives of Corollary 3.2. The F_i span $E_1(T)$ and have just one linear relation, the sum in (4.5), because the same statements hold for the vertex vectors under J.

Fix v_1 and let $G \cong S_n$ be its stabilizer, permuting $v_2, \ldots, v_N$. The function $f = F_1$ is G-invariant. The fixed vectors of G in $\mathbb{R}^n$ form exactly $\mathbb{R}v_1$: being fixed forces orthogonality

to every $v_i - v_j$ with $i, j > 1$, and these differences span the $(n-1)$-dimensional hyperplane perpendicular to v_1. Therefore

$$E_1(T)^G = \mathbb{R}f. \tag{4.6}$$

4.2. **The orthogonal barycenter chain.** The barycentric coordinates on $\overline{T}$ are

$$\lambda_i(x) = \frac{1 + nx \cdot v_i}{N}, \qquad x = \sum_i \lambda_i(x) v_i, \qquad \sum_i \lambda_i(x) = 1. \tag{4.7}$$

Consider the open chamber

$$C = \{x \in T : \lambda_2(x) > \lambda_3(x) > \cdots > \lambda_N(x)\}. \tag{4.8}$$

Its G-images tile T up to their common walls: sorting the distinct coordinates $\lambda_2, \ldots, \lambda_N$ determines a unique chamber. The exceptional set lies in the finitely many hyperplanes $\lambda_i = \lambda_j$, $2 \le i < j \le N$, and has measure zero.

Proposition 4.2 (Orthoscheme geometry)**.** *Define*

$$q_j = \frac{v_2 + \cdots + v_{j+1}}{j} \quad (1 \le j \le n), \qquad r_0 = v_1, \quad r_i = q_{N-i} \quad (1 \le i \le n). \tag{4.9}$$

Then $\overline{C} = \operatorname{conv}\{r_0, \ldots, r_n\}$, and its successive edge vectors $d_i = r_i - r_{i-1}$ are pairwise orthogonal. With $L_i = |d_i|$ and $e_i = d_i/L_i$,

$$e_1 = -v_1, \qquad L_1 = \frac{N}{n}, \qquad L_i^2 = \frac{N}{n(N-i)(N-i+1)} \quad (2 \le i \le n). \tag{4.10}$$

In the orthonormal coordinates

$$y_i = (x - r_0) \cdot e_i, \qquad t_i = y_i / L_i, \tag{4.11}$$

the chamber and its facet-normal directions are

$$C = \{1 > t_1 > t_2 > \cdots > t_n > 0\}, \tag{4.12}$$

$$\nu \parallel e_1, \quad -e_n, \quad -\frac{e_i}{L_i} + \frac{e_{i+1}}{L_{i+1}} (1 \le i < n). \tag{4.13}$$

Proof. First, $\overline{C}$ is precisely the region in $\overline{T}$ given by the weak chamber inequalities. One inclusion follows by continuity. For the other, choose $z \in C$, which exists by prescribing positive barycentric coordinates with $\lambda_2 > \cdots > \lambda_N$. If x satisfies the weak inequalities, then $(1-\varepsilon)x + \varepsilon z \in C$ for $0 < \varepsilon < 1$, and these points tend to x.

Set $\lambda_{N+1} = 0$. If the barycentric coordinates satisfy the weak chamber inequalities, let $b_j = j(\lambda_{j+1} - \lambda_{j+2})$ for $1 \le j \le n$. Then $b_j \ge 0$, and telescoping gives

$$x = \lambda_1 v_1 + \sum_{j=1}^{n} b_j q_j, \qquad \lambda_1 + \sum_{j=1}^{n} b_j = 1.$$

Conversely, $v_1, q_1, \ldots, q_n$ satisfy the weak chamber inequalities, so their convex hull is contained in $\overline{C}$. This proves the convex-hull description. Strict positivity of these $n+1$ convex coefficients is equivalent to $x \in C$.

Direct use of (4.1) gives

$$v_1 \cdot q_j = -\frac{1}{n}, \qquad q_j \cdot q_k = \frac{N-k}{nk} = |q_k|^2 \quad (j \le k). \tag{4.14}$$

The first edge is $q_n - v_1 = -(N/n)v_1$. It is orthogonal to all the subsequent barycenter differences. For $1 \le m < k < n$, (4.14) gives

$$(q_m - q_{m+1}) \cdot q_k = (q_m - q_{m+1}) \cdot q_{k+1} = 0.$$

Thus distinct successive barycenter differences are orthogonal. For $1 \le m < n$,

$$|q_m - q_{m+1}|^2 = |q_m|^2 - |q_{m+1}|^2 = \frac{N}{nm(m+1)}.$$

This proves orthogonality and (4.10).

In the resulting orthonormal coordinates, $r_j = r_0 + \sum_{i=1}^{j} L_i e_i$. The convex hull of these points is described by $1 \ge t_1 \ge \cdots \ge t_n \ge 0$: its convex coefficients at $r_0, r_1, \ldots, r_n$ are $1 - t_1, t_1 - t_2, \ldots, t_{n-1} - t_n, t_n$. Taking the interior proves (4.12). Differentiating its affine defining inequalities in the orthonormal y coordinates gives (4.13). □

The word *orthoscheme* refers to the pairwise orthogonality of the successive edge vectors in (4.9). In every derivative, norm, and energy below, the variables are the orthonormal y_i. The scaled variables t_i are used only to express the chamber inequalities. In particular, the factors L_i^{-1} in (4.13) remain in every Neumann relation.

4.3. **Reflection extension and chamber simplicity.**

Proposition 4.3 (First chamber eigenvalue)**.** *The first positive Neumann eigenvalue of C equals $\mu_1(T)$, and its eigenspace is simple:*

$$\mu_1(C) = \mu_1(T) = \mu, \qquad E_1(C) = \mathbb{R}(f|_C). \tag{4.15}$$

Proof. For $h \in H^1(C)$, define its extension on a chamber gC, $g \in G$, by $\widetilde{h}(gx) = h(x)$ for $x \in C$. Adjacent chambers meet along a wall which is fixed pointwise by their gluing reflection. Their transported traces therefore agree. To verify the global Sobolev regularity, take $\eta \in C_c^\infty(T)$ and integrate by parts on each chamber. Internal facet terms cancel because the traces match and the outward normals oppose; exterior facet terms vanish because η is supported inside T. Hence, for $j = 1, \ldots, n$,

$$\int_T \widetilde{h}\, \partial_j \eta \, dx = -\sum_{g \in G} \int_{gC} \left[g \nabla h(g^{-1}x)\right]_j \eta(x)\, dx.$$

The piecewise gradient on the right lies in $L^2(T)$, proving $\widetilde{h} \in H^1(T)$. Lower-dimensional intersections do not enter the facet integrals and require no additional matching condition. The extension is G-invariant, and

$$\int_T \widetilde{h} = n! \int_C h, \qquad \|\widetilde{h}\|^2_{L^2(T)} = n! \|h\|^2_{L^2(C)}, \qquad \|\nabla \widetilde{h}\|^2_{L^2(T)} = n! \|\nabla h\|^2_{L^2(C)}. \tag{4.16}$$

Applying the Rayleigh principle to a mean-zero chamber minimizer gives $\mu_1(T) \le \mu_1(C)$.

Conversely, f and f^2 are G-invariant, as is $|\nabla f|^2$. Thus $\int_C f = 0$ and $\|f\|^2_{L^2(C)} = \|f\|^2_{L^2(T)}/n! > 0$, while the Rayleigh quotient of $f|_C$ equals μ. The chamber Rayleigh principle gives the reverse inequality. Equality in (2.1), applied to the convex polytope C, now gives $f|_C \in E_1(C)$. In particular, its weak Neumann condition holds on every chamber facet, including the artificial reflection walls.

If $0 \ne h \in E_1(C)$, its extension has zero mean and global Rayleigh quotient μ. The equality case of (2.1) on T puts it in $E_1(T)^G$, which is $\mathbb{R}f$ by (4.6). Hence h is a multiple of $f|_C$. Since $f|_C$ is a nonzero member of $E_1(C)$, this proves both claims. □

5. Chamber monotonicity and the extrema of vertex modes

Retain the vertex mode $f = F_1$ and the chamber C with the orthonormal basis $e_1, \ldots, e_n$ constructed in Section 4. In this section f also denotes its restriction to C. By Proposition 4.3 it is a first Neumann eigenfunction there, and convexity gives $f \in H^2(C)$.

Proposition 5.1 (Strict signs of all chamber derivatives)**.** *For every $i = 1, \dots, n$,*

$$D_{e_i} f < 0 \quad \text{throughout } C. \tag{5.1}$$

Proof. Write $f_i = D_{e_i} f \in H^1(C)$. The facet normals in (4.13) give the trace conditions

$$f_1 = 0 \quad \text{on } t_1 = 1, \qquad f_n = 0 \quad \text{on } t_n = 0, \tag{5.2}$$

$$\frac{f_i}{L_i} = \frac{f_{i+1}}{L_{i+1}} \quad \text{on } t_i = t_{i+1} \quad (1 \le i < n). \tag{5.3}$$

Since every $L_i > 0$, these relations are preserved by simultaneous replacement of f_i with $-|f_i|$. Thus

$$X = -\sum_{i=1}^{n} |f_i| e_i \in \mathcal{V}(C). \tag{5.4}$$

The trace operation commutes with Lipschitz composition, so this assertion applies to the H^1 traces on all facets.

The Sobolev chain rule gives

$$|X|^2 = |\nabla f|^2, \qquad |DX|^2 = |D^2 f|^2 \quad \text{almost everywhere in } C. \tag{5.5}$$

In the second equality one uses that the weak gradient of a Sobolev function is zero almost everywhere on its zero set; hence taking its absolute value preserves the squared gradient norm also there. Corollary 2.2 and Proposition 4.3 yield

$$\|DX\|^2_{L^2(C)} = \|D^2 f\|^2_{L^2(C)} = \mu \|\nabla f\|^2_{L^2(C)} = \mu \|X\|^2_{L^2(C)}.$$

The equality case of Proposition 2.4, applied to C, therefore gives $X = \nabla h$ for $h \in E_1(C)$. Chamber simplicity implies $h = cf$ for a real constant c. Since f is nonconstant, (5.5) gives $c^2 = 1$.

The vector $e_1 = -v_1$ is fixed by G. Equivariance of the gradient of the G-invariant function f shows that f_1 is also G-invariant. Consequently,

$$\int_C f_1 \, dx = \frac{1}{n!} \int_T D_{e_1} f \, dx = \frac{1}{n!} J(f) \cdot e_1 = -\frac{1}{n!} < 0. \tag{5.6}$$

Also, $\int_C X \cdot e_1 = -\int_C |f_1| < 0$. Since $X = c\nabla f$, the signs of these integrals force $c = 1$. Hence

$$f_i \le 0 \quad \text{in } C \qquad (1 \le i \le n). \tag{5.7}$$

Interior elliptic regularity makes f smooth in C, so these inequalities hold pointwise and the eigenvalue equation can be differentiated there.

Each $g_i = -f_i$ is nonnegative and satisfies $-\Delta g_i = \mu g_i$ in the interior. In particular, $\Delta g_i \le 0$. The strong minimum principle gives the alternative $f_i < 0$ throughout C or $f_i \equiv 0$. The first component is not identically zero by (5.6), so $f_1 < 0$.

We prove $f_j < 0$ successively for $j = 2, \dots, n$. Suppose $f_{j-1} < 0$ and, contrary to the claim, $f_j \equiv 0$. Choose q in the relative interior of the facet $\{t_{j-1} = t_j\} \cap \partial C$ and set

$$w = L_{j-1} e_{j-1} + L_j e_j, \qquad g = -D_w f = -L_{j-1} f_{j-1} > 0 \quad \text{in } C.$$

The facet-normal formula (4.13) gives $w \cdot \nu = 0$ on this facet, the only one incident to q. Lemma 3.3, applied to the Neumann eigenfunction f on C, gives the local weak Neumann equation for g. Lemma 3.1 then gives a continuous representative near q with $g(q) > 0$. On the other hand, (5.3) and $f_j \equiv 0$ give zero trace for f_{j-1}, and hence for g, on this facet. The continuous representative agrees with its Sobolev trace there, so its zero trace and continuity imply $g(q) = 0$. This contradiction excludes $f_j \equiv 0$; the strong minimum principle alternative therefore gives $f_j < 0$. Induction proves (5.1). □

Proposition 5.2 (Exact extrema of a vertex mode)**.** *For each $i = 1, \dots, N$,*

$$\arg\max_{\overline{T}} F_i = \{v_i\}, \qquad \arg\min_{\overline{T}} F_i = \{v_j : j \neq i\}. \tag{5.8}$$

Every nonvertex point of $\overline{T}$ has a value strictly between the minimum and maximum of F_i.

Proof. It suffices to treat $f = F_1$. Whenever distinct $p_0, p_1 \in \overline{C}$ satisfy $p_1 - p_0 = \sum_i a_i e_i$ with $a_i \geq 0$, at least one coefficient is positive. Proposition 5.1 gives $D_{p_1 - p_0} f < 0$ throughout C. Lemma 3.4, applied to the Neumann eigenfunction on C, therefore gives $f(p_1) < f(p_0)$, including when the segment lies in the boundary.

For $x \in \overline{C}$, Proposition 4.2 gives

$$x - r_0 = \sum_{i=1}^{n} L_i t_i(x) e_i, \qquad r_n - x = \sum_{i=1}^{n} L_i (1 - t_i(x)) e_i.$$

Both sets of coefficients are nonnegative. All coefficients in the first expression vanish exactly when $x = r_0$; all in the second vanish exactly when $x = r_n$. Applying the comparison to (r_0, x) and (x, r_n) yields $f(x) < f(r_0)$ for $x \neq r_0$ and $f(r_n) < f(x)$ for $x \neq r_n$. Thus the unique maximum on $\overline{C}$ is $r_0 = v_1$, and the unique minimum is $r_n = v_2$, including all lower-dimensional boundary strata.

The G-images of $\overline{C}$ cover $\overline{T}$; G fixes v_1 and acts transitively on $\{v_2, \dots, v_N\}$. By invariance of f, the chamber extrema therefore give $\arg\max_{\overline{T}} f = \{v_1\}$ and $\arg\min_{\overline{T}} f = \{v_2, \dots, v_N\}$. Every nonvertex lies in a closed chamber and differs from both of its extreme vertices, so both inequalities are strict there. Equivariance under all vertex permutations proves the assertion for each F_i. □

6. Positive interpolation and proof of the main theorems

We use the continuous representatives supplied by Corollary 3.2. Thus identities in $\mathbb{R} \oplus E_1(T)$ hold on $\overline{T}$ whenever they hold in $H^1(T)$. By (4.5), $F_{\sigma(i)}(v_{\sigma(j)}) = F_i(v_j)$ for every vertex permutation σ. Such permutations are transitive on diagonal ordered pairs and on off-diagonal ordered pairs, giving two common numbers α, β such that

$$F_i(v_i) = \alpha, \qquad F_i(v_j) = \beta \quad (i \neq j). \tag{6.1}$$

Proposition 5.2 implies $\alpha > \beta$. Evaluating $\sum_i F_i = 0$ at any vertex gives $\alpha + n\beta = 0$. Put

$$\kappa = \alpha - \beta > 0, \qquad \alpha = \frac{n}{N}\kappa, \qquad \beta = -\frac{1}{N}\kappa, \qquad \psi_i = \frac{F_i}{\kappa}, \quad h_i = \psi_i + \frac{1}{N}. \tag{6.2}$$

Proposition 6.1 (Cardinal values and positive weights)**.** *The functions in (6.2) satisfy*

$$\psi_i(v_j) = \delta_{ij} - \frac{1}{N}, \qquad \sum_{i=1}^{N} \psi_i = 0, \tag{6.3}$$

$$h_i(v_j) = \delta_{ij}, \qquad \sum_{i=1}^{N} h_i = 1. \tag{6.4}$$

At every nonvertex point $x \in \overline{T}$ one has $0 < h_i(x) < 1$ for each i. The ψ_i span $E_1(T)$ with the sole linear relation $\sum_i \psi_i = 0$.

Proof. The identities follow directly from (6.1)–(6.2) and $\sum_i F_i = 0$. At a nonvertex, Proposition 5.2 gives $\beta < F_i(x) < \alpha$. Since $h_i = (F_i - \beta)/\kappa$ and $\alpha - \beta = \kappa > 0$, these inequalities give $0 < h_i(x) < 1$. The assertions about the span and its only relation follow from the same assertions for the F_i in Section 4. □

Proof of Theorems 1.1 and 1.2. Continuity is Corollary 3.2, and the dimension is Proposition 4.1. Every $u \in E_1(T)$ can be written as a linear combination of the ψ_i. Subtracting the mean of its coefficients, using $\sum_i \psi_i = 0$, gives a representation

$$u = \sum_{i=1}^{N} c_i \psi_i, \qquad \sum_i c_i = 0. \tag{6.5}$$

Evaluating (6.5) at a vertex using (6.3) gives

$$u(v_j) = \sum_i c_i(\delta_{ij} - 1/N) = c_j. \tag{6.6}$$

Thus the coefficients are uniquely determined by the vertex values, and those values have zero sum. In particular, vanishing vertex values force $u = 0$, so vertex evaluation is injective. Conversely, any zero-sum $c \in \mathbb{R}^N$ defines an eigenfunction by (6.5) with exactly those vertex values. This proves (1.4).

The functions h_i of Proposition 6.1 give the existence and positivity assertions in Theorem 1.2. To check uniqueness in $\mathbb{R} \oplus E_1(T)$, suppose $a + u$ vanishes at every vertex, with $a \in \mathbb{R}$ and $u \in E_1(T)$. By (6.6), $\sum_i u(v_i) = 0$. Summing the equations $a + u(v_i) = 0$ gives $Na = 0$. Thus $a = 0$, and injectivity of vertex evaluation gives $u = 0$. Applying this fact to the difference of any two cardinal functions proves uniqueness. Since ψ_i is a Neumann eigenfunction and $h_i = \psi_i + 1/N$, for every $\eta \in H^1(T)$ one has

$$\int_T \nabla h_i \cdot \nabla \eta \, dx = \mu \int_T \psi_i \eta \, dx = \mu \int_T (h_i - 1/N)\eta \, dx.$$

This is precisely the weak formulation of (1.11), including the homogeneous Neumann condition.

Using $\sum_i c_i = 0$ in (6.5) yields

$$u(x) = \sum_{i=1}^{N} c_i h_i(x) = \sum_{i=1}^{N} u(v_i) h_i(x). \tag{6.7}$$

This proves the interpolation formula in Theorem 1.2.

Let $u \neq 0$, and put $m = \min_i c_i$ and $M = \max_i c_i$. The zero-sum data are not all equal, by injectivity, so $m < M$. For a nonvertex $x \in \overline{T}$, all weights in (6.7) are positive and sum to one. Hence

$$u(x) - m = \sum_i (c_i - m) h_i(x) > 0, \qquad M - u(x) = \sum_i (M - c_i) h_i(x) > 0. \tag{6.8}$$

Each sum has nonnegative summands and at least one strictly positive summand. Thus all nonvertices have strictly intermediate values, and the maximum and minimum sets are exactly the vertices where $c_i = M$ and $c_i = m$. This proves (1.5)–(1.7), including every tie. Finally, the extremal-set description gives $|u(x)| \leq \max_i |c_i|$ on $\overline{T}$. Choose k with $|c_k| = \max_i |c_i|$. For each $\varepsilon > 0$, continuity gives $r > 0$ such that

$$|u(x)| > \max_i |c_i| - \varepsilon \qquad (x \in T \cap B_r(v_k)).$$

This intersection is nonempty and open, hence has positive volume. Consequently the essential supremum is at least $\max_i |c_i| - \varepsilon$. Letting $\varepsilon \downarrow 0$ proves (1.8); its assertion for $u = 0$ is immediate. Both theorems are proved. □

Corollary 6.2 (An explicit choice of basis)**.** *Any n of the functions $\psi_1, \dots, \psi_N$ form a basis of $E_1(T)$. In particular,*

$$E_1(T) = \operatorname{span}\{\psi_1, \dots, \psi_n\}, \qquad u = \sum_{i=1}^{n} (u(v_i) - u(v_N))\psi_i. \tag{6.9}$$

Proof. A relation among n generators extends to a relation among all N with one zero coefficient. Since their only relation has all coefficients equal, every coefficient must be zero. They are thus independent and form a basis of the n-dimensional eigenspace. The representation follows from (6.5) by substituting $\psi_N = -\sum_{i=1}^{n} \psi_i$. □

7. Complete extremal-set classification and illustrations

Throughout this section, $N = n + 1$, $n \geq 2$, and the canonical generators $\psi_1, \dots, \psi_N$ are normalized as in Theorem 1.2. Thus every $u \in E_1(T)$ has the unique representation

$$u = \sum_{i=1}^{N} c_i \psi_i, \qquad c_i = u(v_i), \qquad \sum_{i=1}^{N} c_i = 0. \tag{7.1}$$

The following consequences describe every possible pattern of global maximum and minimum points, including all ties.

Corollary 7.1 (All extremal-set patterns)**.** *Let $u \in E_1(T) \setminus \{0\}$, and define*

$$A = \{i : c_i = \max_j c_j\}, \qquad B = \{i : c_i = \min_j c_j\}, \qquad R = \{1, \dots, N\} \setminus (A \cup B).$$

Then A and B are nonempty and disjoint, and

$$\arg\max_{\overline{T}} u = \{v_i : i \in A\}, \qquad \arg\min_{\overline{T}} u = \{v_j : j \in B\}. \tag{7.2}$$

Every nonvertex point of $\overline{T}$ has a value strictly between the global minimum and the global maximum.

Conversely, every pair of disjoint nonempty subsets $A, B \subset \{1, \dots, N\}$ occurs in (7.2) for some $u \in E_1(T) \setminus \{0\}$. In particular, the possible numbers p and q of maximum and minimum points are exactly

$$p \geq 1, \qquad q \geq 1, \qquad p + q \leq N. \tag{7.3}$$

For fixed A, B, with $|A| = p$ and $|B| = q$, the functions having these exact extremal sets form a nonempty convex cone with the zero function omitted, of dimension $N + 1 - p - q$, relatively open in its linear span in $E_1(T)$.

Proof. For nonzero u, Theorem 1.1 gives distinct minimum and maximum values and identifies their vertex sets. This proves all the direct assertions.

For the converse, put $p = |A|$ and $q = |B|$ and prescribe

$$c_i = \begin{cases} q, & i \in A, \\ -p, & i \in B, \\ 0, & i \in R. \end{cases} \tag{7.4}$$

These data have zero sum and are nonzero. The vertex-evaluation isomorphism gives a unique eigenfunction with these values. Since $-p < 0 < q$, its exact maximum and minimum index sets are A and B. This also proves the sufficiency of all the cardinality conditions in (7.3).

To verify the final assertion, write $r = |R| = N - p - q$. In vertex coordinates the indicated set is described by real variables $M, m, (c_\ell)_{\ell \in R}$ satisfying

$$pM + qm + \sum_{\ell \in R} c_\ell = 0, \qquad m < M, \qquad m < c_\ell < M \quad (\ell \in R),$$

with $c_i = M$ on A and $c_j = m$ on B. The linear equation defines an $(r+1)$-dimensional hyperplane in $r+2$ variables, including when $r = 0$. Assigning M on A, m on B, and c_ℓ on R identifies this hyperplane linearly and bijectively with the subspace of H_0 consisting of data constant on each of the two blocks A, B. Here injectivity uses that both blocks are nonempty. The strict inequalities define a relatively open convex subset of this subspace, invariant under positive scaling and excluding zero. It is nonempty by (7.4). A nonempty relatively open subset of a linear space spans that space, so its linear span has dimension $r+1 = N+1-p-q$. The vertex-evaluation isomorphism of Theorem 1.1 transfers all these properties to $E_1(T)$. □

Thus a unique maximum and a unique minimum correspond to $p = q = 1$. Ties of p maximum vertices and q minimum vertices impose exactly $(p-1)+(q-1)$ independent linear equalities on the n-dimensional eigenspace: the block-constant subspace just described has codimension $(N-1)-(N+1-p-q) = p+q-2$. The remaining vertices, when present, have intermediate values. Multiplication of an eigenfunction by a negative scalar interchanges its maximum and minimum sets.

Corollary 7.2 (Subset modes and the two-level cases)**.** *For a nonempty proper subset $A \subset \{1,\dots,N\}$, let $k = |A|$ and set*

$$U_A = \sum_{i\in A} \psi_i. \tag{7.5}$$

Then

$$U_A(v_j) = \begin{cases} 1-k/N, & j \in A, \\ -k/N, & j \notin A, \end{cases} \tag{7.6}$$

and

$$\arg\max_{\overline{T}} U_A = \{v_j : j \in A\}, \qquad \arg\min_{\overline{T}} U_A = \{v_j : j \notin A\}. \tag{7.7}$$

Every nonzero first eigenfunction for which every vertex is a global maximum or a global minimum is a positive multiple of exactly one U_A. More precisely, if its maximum and minimum values are M and m and its maximum index set is A, then

$$u = (M-m)U_A. \tag{7.8}$$

Also, $U_{A^c} = -U_A$.

Proof. The vertex normalization of the ψ_i gives (7.6) by summation. The two displayed vertex values differ by one, so $U_A \neq 0$, and Corollary 7.1 gives (7.7).

For the converse, the zero sum of all vertex values gives $kM + (N-k)m = 0$. Together with $M - m > 0$, this implies

$$M = (M-m)(1-k/N), \qquad m = -(M-m)k/N.$$

Hence u and $(M-m)U_A$ have the same vertex values. Injectivity of vertex evaluation proves (7.8). The maximum index set determines A uniquely. Finally, $U_A + U_{A^c} = \sum_i \psi_i = 0$. □

Corollary 7.3 (No extremal face plateaux)**.** *Let $u \in E_1(T) \setminus \{0\}$ and let F be a positive-dimensional face of $\overline{T}$, allowing $F = \overline{T}$. At every nonvertex point $x \in F$,*

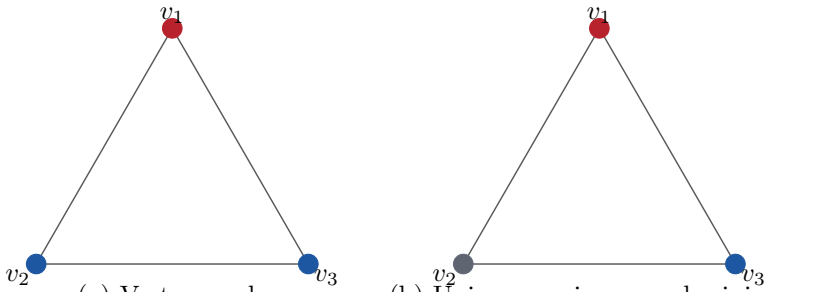
$$\min_{\overline{T}} u < u(x) < \max_{\overline{T}} u.$$

In particular, even when all vertices of F share the global maximum or all share the global minimum, this value is attained on F exactly at those vertices.

Proof. Every such x is a nonvertex point of $\overline{T}$, so the assertion is precisely the strict part of Theorem 1.1. □

7.1. **Illustrations in dimensions two, three, and four.** The following diagrams use *red* markers for global maximum points, *blue* markers for global minimum points, and *gray* markers for vertices with intermediate values. Only the red and blue vertex markers indicate extrema. Edges and faces are not marked as extremal sets, in accordance with Corollary 7.3. In each diagram the vector c lists the values $(u(v_1), \dots, u(v_N))$ and uniquely specifies the eigenfunction by (7.1).

For an equilateral triangle, the possibilities for the numbers of maximum and minimum points are $(p, q) = (1, 1), (1, 2), (2, 1)$. The first two are shown in Figure 1; changing the sign of a vertex mode gives the third.



(a) Vertex mode $c = (2, -1, -1)$

(b) Unique maximum and minimum $c = (1, 0, -1)$

FIGURE 1. Two extremal-set patterns on the equilateral triangle. In (a), $u = 3\psi_1$ has its unique maximum at v_1 and its two minimum points at v_2, v_3. In (b), $u = \psi_1 - \psi_3$ has its unique maximum at v_1, its unique minimum at v_3, and the intermediate vertex value $u(v_2) = 0$. The interior of the blue-ended edge in (a) has values strictly greater than the global minimum.

For a regular tetrahedron, the possibilities are

$$(p, q) \in \{(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (2, 2)\}.$$

Figure 2 illustrates a vertex mode and the balanced two-level case $(p, q) = (2, 2)$. All patterns with intermediate vertices follow from (7.4).

For a regular 4-simplex, the zero-sum data

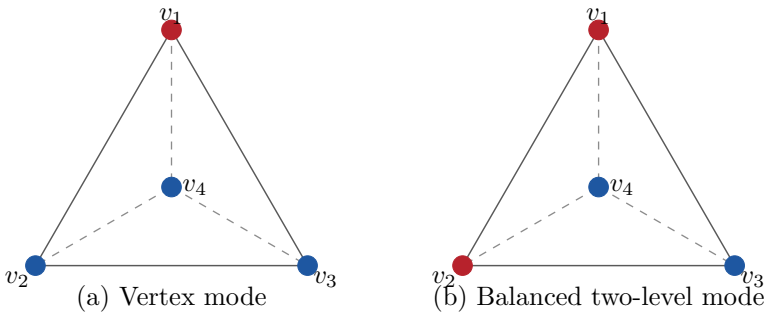
$$c = (2, 2, 0, -2, -2) \tag{7.9}$$

give an example with two maximum points, two minimum points, and one intermediate vertex. The eigenfunction is $u = 2(\psi_1 + \psi_2 - \psi_4 - \psi_5)$. Figure 3 uses a planar orthogonal projection in which the five vertices form a regular pentagon. To verify that this projection exists, realize the simplex in $\{x \in \mathbb{R}^5 : \sum_i x_i = 0\}$ with vertices

$$v_j = \sqrt{\frac{5}{4}} \left(e_j - \frac{1}{5}\mathbf{1} \right), \qquad j = 1, \dots, 5.$$

Here e_j denotes the jth standard coordinate vector of $\mathbb{R}^5$, and $\mathbf{1} = (1, 1, 1, 1, 1)$. These vertices have unit norm and satisfy $v_i \cdot v_j = -1/4$ for $i \neq j$, so they form a regular 4-simplex with the normalization (4.1). Set $\theta_j = \pi/2 - 2\pi(j-1)/5$. The two vectors

$$a = \sqrt{\frac{2}{5}} (\cos \theta_j)_{j=1}^5, \qquad b = \sqrt{\frac{2}{5}} (\sin \theta_j)_{j=1}^5$$



(a) Vertex mode $c = (3, -1, -1, -1)$ (b) Balanced two-level mode $c = (1, 1, -1, -1)$

FIGURE 2. Orthographic projections of a regular tetrahedron. The vertex v_4 is behind the face with vertices v_1, v_2, v_3; its projected image is the centroid of the outer triangle, and its three incident edges are dashed. In (a), $u = 4\psi_1$ has one maximum point and three minimum points. In (b), $u = 2U_{\{1,2\}}$ has precisely the two red vertices as maxima and the two blue vertices as minima. In particular, the segments joining equal-colored vertices are not extremal plateaux.

lie in that hyperplane and are orthonormal. Indeed, the fifth-root-of-unity sums give

$$\sum_{j=1}^{5} e^{\mathrm{i}\theta_j} = \sum_{j=1}^{5} e^{2\mathrm{i}\theta_j} = 0.$$

Their real and imaginary parts, together with the double-angle identities, yield

$$\sum_j \cos\theta_j = \sum_j \sin\theta_j = \sum_j \cos\theta_j \sin\theta_j = 0,$$

$$\sum_j \cos^2\theta_j = \sum_j \sin^2\theta_j = \frac{5}{2}.$$

Thus $a \cdot \mathbf{1} = b \cdot \mathbf{1} = a \cdot b = 0$ and $|a| = |b| = 1$. In the orthonormal coordinates (a, b) of this plane, the projection of v_j has coordinates

$$(v_j \cdot a, v_j \cdot b) = 2^{-1/2}(\cos\theta_j, \sin\theta_j),$$

where the factor is $\sqrt{5/4}\sqrt{2/5} = 2^{-1/2}$. A common rescaling therefore gives the diagram. The projected edge crossings carry no additional vertex labels and do not represent additional extrema.

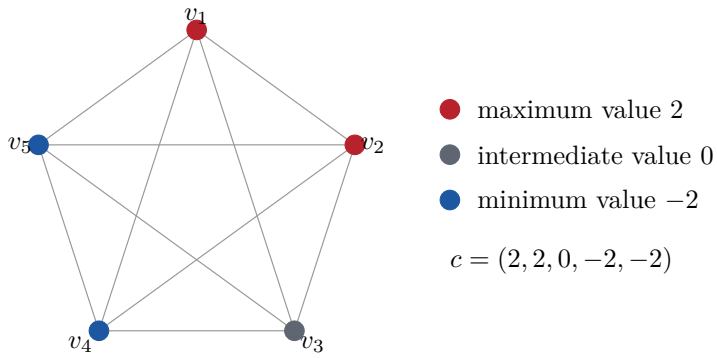


FIGURE 3. A planar orthogonal projection of a regular 4-simplex, showing all ten edges and the eigenfunction with vertex data (7.9). Its exact maximum set is $\{v_1, v_2\}$, its exact minimum set is $\{v_4, v_5\}$, and $u(v_3) = 0$. The pentagon and its crossing segments are a projection of a four-dimensional domain. Only the five labeled markers represent simplex vertices.

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